CBS311 Fundamentals of Data Science

Predicting House Prices Using USA Housing Data

Project

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1. Introduction

In today's real estate market, estimating the price of a house accurately is a key challenge.
Machine learning techniques can help predict house prices based on various factors such as income, area population, and number of rooms.
This project uses the USA Housing dataset to develop a regression model that predicts house prices using these influencing features.

2. Objective

The main objective of this project is to **predict house prices** based on different features of the housing area.
This is a **regression problem**, where the target variable is the continuous numeric value **Price**.

3. Dataset Information

- **Dataset Name:** USA_Housing.csv
- **Source:** Kaggle (USA Housing Dataset)
- **Number of Records:** ~5000
- **Number of Features:** 7 (including target variable)

Feature Name	Description
Avg. Area Income	Average income of people living in the area
Avg. Area House Age	Average age of houses in the area
Avg. Area Number of Rooms	Average number of rooms per house
Avg. Area Number of Bedrooms	Average number of bedrooms
Area Population	Total population in the area
Address	Address of the house (removed during modeling)
Price	Target variable (house price)

4. Methodology

1. **Data Preprocessing:**

- Loaded the dataset using pandas and removed the non-numeric Address column.
- Checked for missing values (none found).
- Split data into 80% training and 20% testing sets.

2. **Exploratory Data Analysis (EDA):**

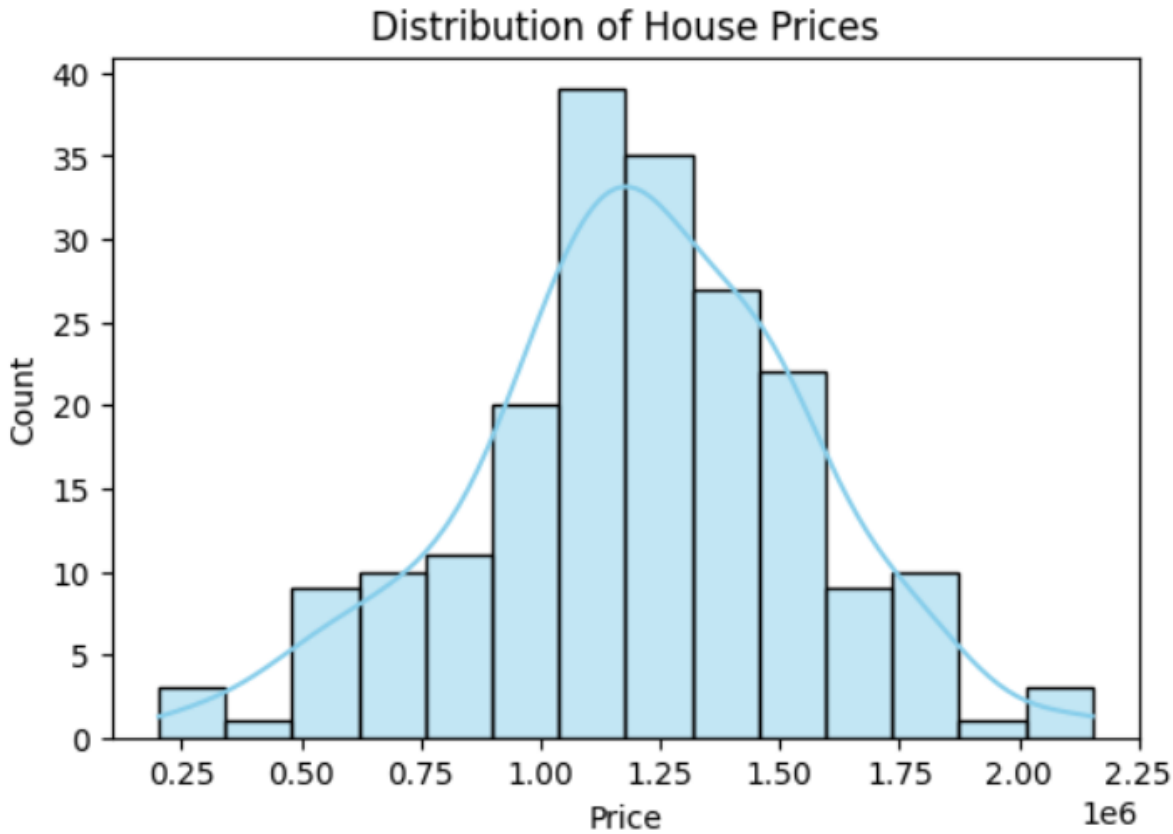
- Created histograms, scatter plots, and a correlation heatmap.
- Observed that house price is highly correlated with Average Area Income and Average Area Number of Rooms.

3. **Model Used:**

- Applied Linear Regression to predict house prices.
- Evaluated performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

5. Results and Observations

- **Histogram:** Shows that most house prices fall in the mid-range with a few high-priced outliers.



- **Correlation Heatmap:** Avg. Area Income and Avg. Area Number of Rooms show strong positive correlation with price.
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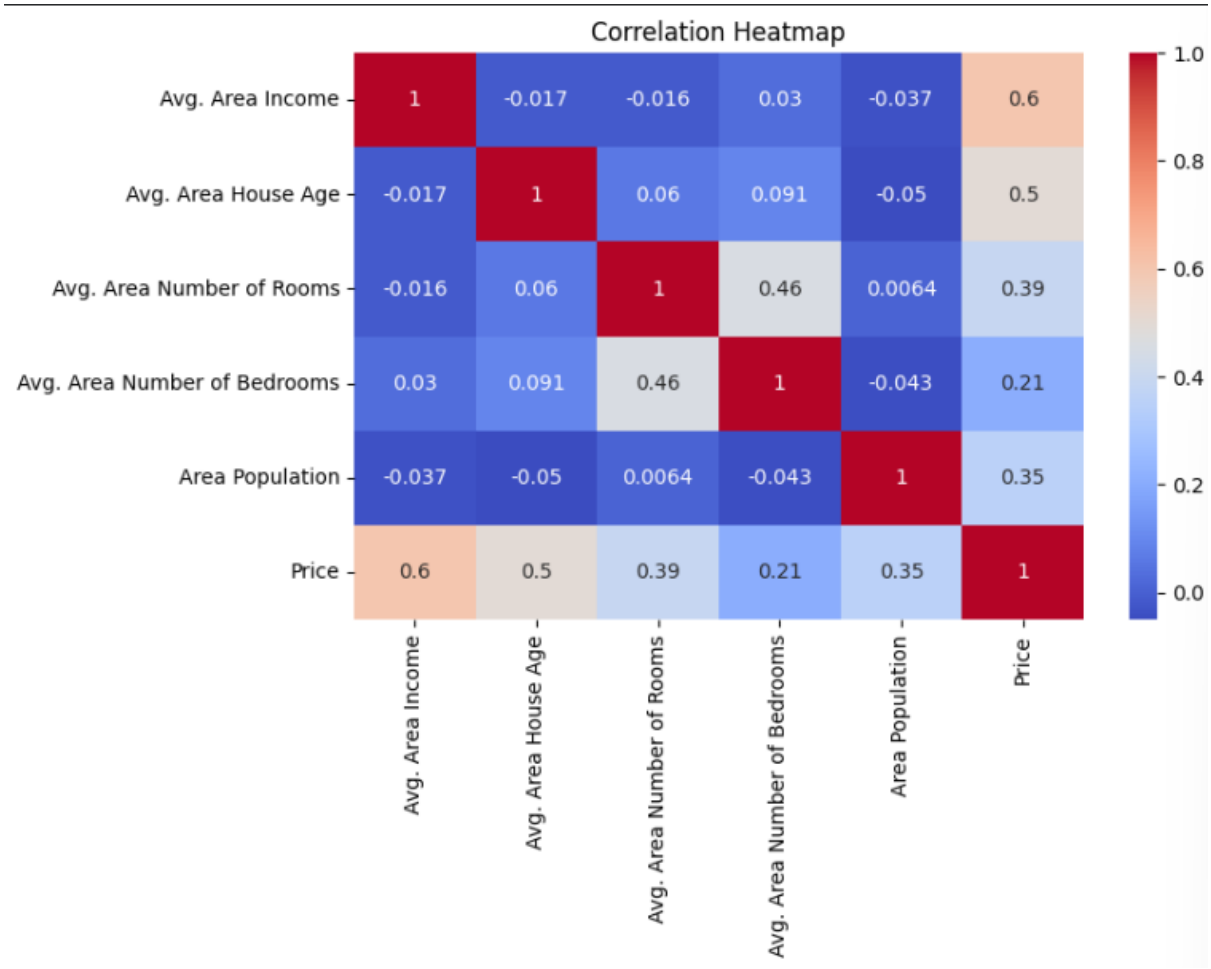
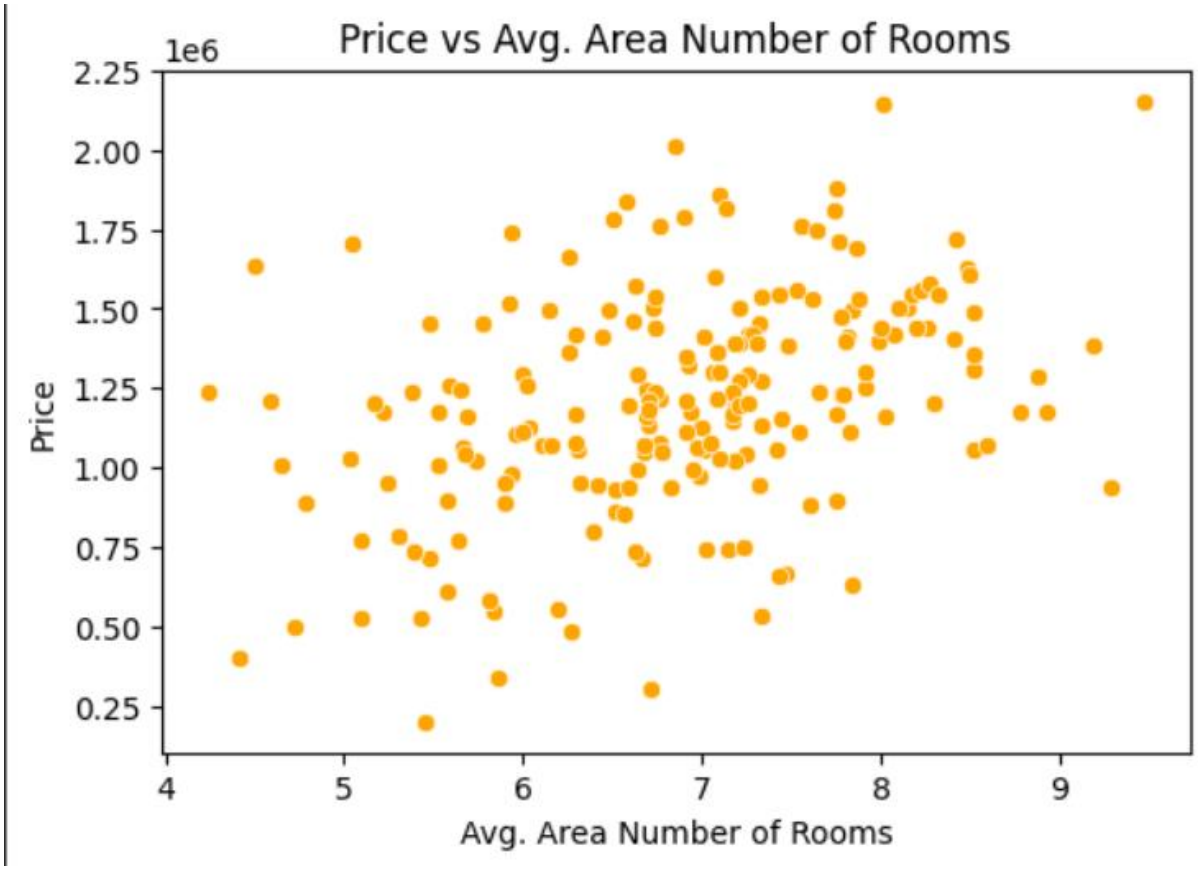
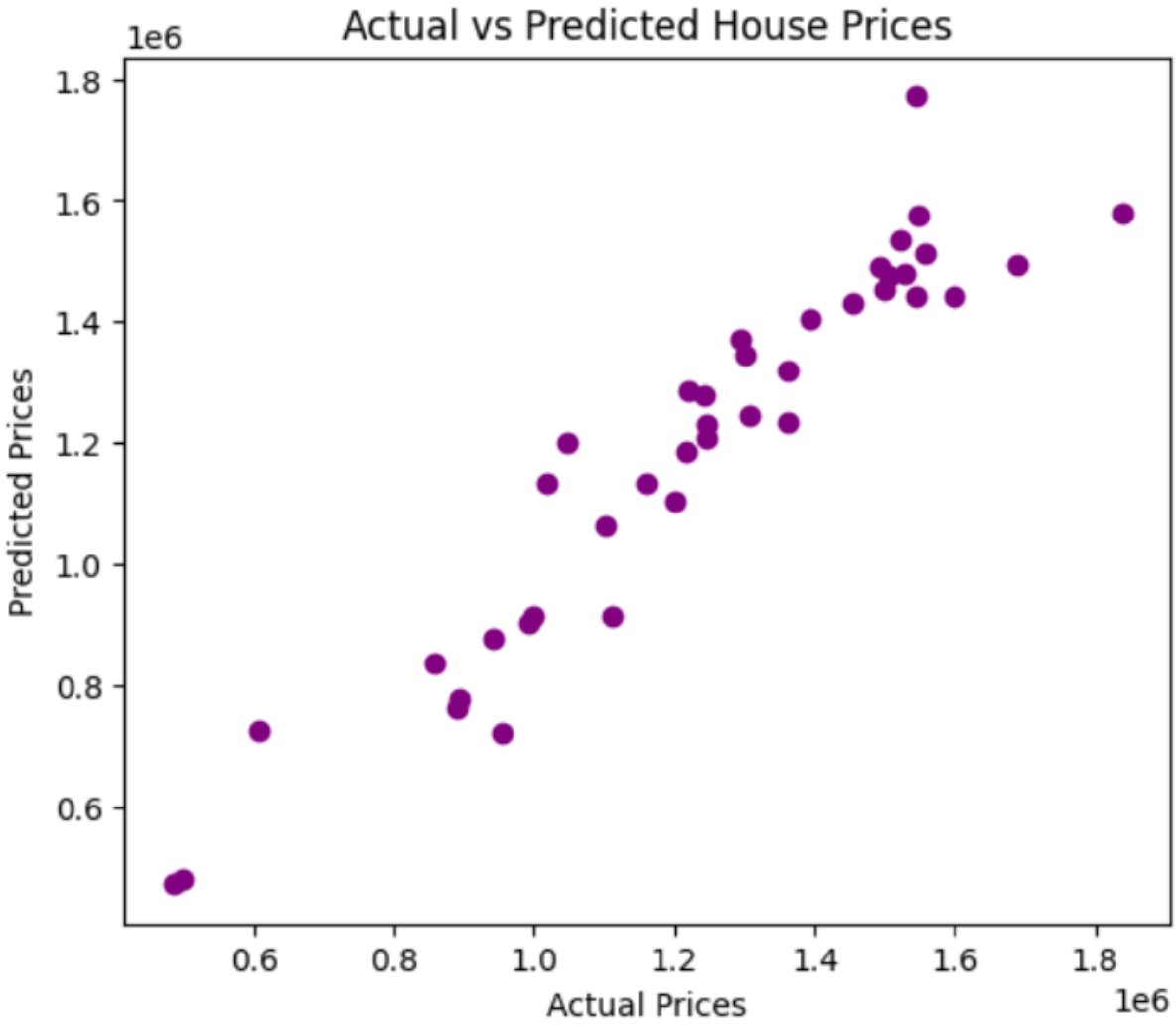

Correlation Heatmap
Avg. Area Income
Avg. Area House Age
Avg. Area Number of Rooms
Avg. Area Number of Bedrooms
Area Population
Price
1
-0.017
-0.016
0.03
-0.037
0.6
-0.017
1
0.06
0.091
-0.05
0.5
-0.016
0.06
1
0.46
0.0064
0.39
0.03
0.091
0.46
1
-0.043
0.21
-0.037
-0.05
0.0064
-0.043
1
0.35
0.6
0.5
0.39
0.21
0.35
1
1.0
0.8
0.6
0.4
0.2
0.0

- **Scatter Plot (Price vs Income):** Higher income areas have higher house prices.

Scatter Plot (Price vs Rooms): Houses with more rooms are generally more expensive.



- **Actual vs Predicted Plot:** Predicted prices closely match actual prices, indicating good model accuracy.



Model Evaluation Results (sample output):

Metric	Value
MAE	~81,000
RMSE	~102,000
R^2 Score	~0.91

These results show that the Linear Regression model explains about **91% of the variance** in house prices.

6. Conclusion

The project successfully built a regression model to predict house prices using the USA Housing dataset.

The analysis revealed that **Average Area Income** and **Average Area Number of Rooms** are the most influential factors in determining house prices.

The Linear Regression model achieved high accuracy, showing that simple regression methods can effectively predict real estate prices.

Future improvements could include using advanced models like **Random Forest** or **Gradient Boosting** for even higher accuracy.